Experiment 27: Online Bus Pass Management System

Aim:

To develop an interactive online bus pass system for students and working people to apply for new bus passes, renew existing passes, purchase e-tickets, manage wallet balance, and obtain QR codes. Administrators can approve passes, manage buses and routes, while conductors verify tickets and update bus locations.

Algorithm:

1. Design a responsive website with separate User, Admin, and Conductor portals.
2. Allow users to register, log in, and enter personal and school or organization details.
3. Provide forms to apply for a new bus pass, renew a pass, and purchase e-tickets.
4. Display bus routes, stops, timings, and estimated travel time.
5. Allow users to add money to a demo e-wallet and view their balance.
6. Allow the administrator to manage buses, routes, conductor accounts, and pass approvals.
7. Generate a QR code for an approved pass or ticket.
8. Allow conductors to scan a QR code, verify the pass or ticket ID, and update bus location.
9. Display the pass status and booking information to the user.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1">

<title>SwiftPass | Online Bus Pass</title>

<script
src="https://cdnjs.cloudflare.com/ajax/libs/qrcodejs/1.0.0/qrcode.min.js"></script>

<style>
```

```
:root{--bg:#f3f6fb;--card:#fff;--ink:#18243a;--muted:#718096;--blue:#245bea;--line:#e2e8f0;--green:#087f5b}

*{box-sizing:border-box}

body{margin:0;background:var(--bg);font-family:Arial,sans-serif;color:var(--ink)}

header{background:linear-gradient(120deg,#102b63,#2865ed);color:white;padding:24px 5%;display:flex;justify-content:space-between;align-items:center;gap:12px;flex-wrap:wrap}

header h1{margin:0;font-size:26px}header p{margin:6px 0 0;color:#dce8ff}

.pill{background:#ffffff24;padding:9px 13px;border-radius:25px;font-size:13px}

nav{display:flex;gap:8px;padding:13px 5%;background:white;overflow-x:auto;position:sticky;top:0;z-index:3;border-bottom:1px solid var(--line)}

nav button{white-space:nowrap;border:0;padding:11px 15px;border-radius:9px;background:#edf2ff;color:#254a9c;cursor:pointer;font-weight:bold}

nav button:hover{background:#dbe6ff}

main{max-width:1100px;margin:auto;padding:25px 5%}

section{margin-bottom:22px}.hidden{display:none!important}

.hero{background:linear-gradient(120deg,#e0eaff,#f8faff);border:1px solid #cddcff;border-radius:20px;padding:27px}

.hero h2{font-size:clamp(25px,5vw,38px);margin:10px 0}

.hero p,.muted{color:var(--muted);line-height:1.6}

button.primary{background:var(--blue);color:white;border:0;padding:12px 16px;border-radius:9px;cursor:pointer;font-weight:bold}

button.secondary{background:#e9efff;color:#254a9c;border:0;padding:10px 13px;border-radius:8px;cursor:pointer}

.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:15px;margin:18px 0}

.card{background:var(--card);padding:20px;border:1px solid var(--line);border-radius:15px;box-shadow:0 5px 18px #243b5308;min-width:0}

.card h3{margin-top:0}

.metric{font-size:29px;font-weight:bold;margin:12px 0}

label{display:block;font-size:13px;font-weight:bold;margin:13px 0 6px}
```

```
input,select{width:100%;padding:12px;border:1px solid #ccd6e3;border-radius:8px;background:#fff;color:var(--ink)}

form button{margin-top:15px}

.row{display:grid;grid-template-columns:1fr 1fr;gap:12px}

table{border-collapse:collapse;width:100%;font-size:13px}

th,td{text-align:left;padding:11px;border-bottom:1px solid var(--line)}

th{background:#f5f8ff;color:#294e9c}

.tablewrap{overflow-x:auto}

.status{display:inline-block;background:#d9f8e9;color:#087f5b;border-radius:15px;padding:5px 9px;font-size:11px;font-weight:bold}

.warning{background:#fff7dd;color:#825b08;padding:12px;border-radius:9px;line-height:1.5;font-size:13px}

#qr{margin:15px auto;display:flex;justify-content:center}

.pass{border:2px dashed #9cb8ff;border-radius:15px;padding:20px;background:#f6f9ff}

footer{text-align:center;padding:25px;color:var(--muted)}

@media(max-width:600px){.row{grid-template-columns:1fr}main{padding:17px 4%}.hero{padding:20px}}

</style>

</head>

<body>

<header>

 <div><h1>🚌 SwiftPass</h1><p>Smart Bus Pass & Ticket Management</p></div>

 <div class="pill">● Demo System</div>

</header>


<nav>

 <button onclick="page('home')">Dashboard</button>

 <button onclick="page('apply')">Apply / Renew</button>

 <button onclick="page('wallet')">My Wallet</button>
```

```
 <button onclick="page('myPass')">My Pass</button>
 <button onclick="page('routes')">Bus Routes</button>
 <button onclick="page('conductor')">Conductor</button>
 <button onclick="page('admin')">Admin</button>
</nav>

<main>
<section id="home">
 <div class="hero">
  <span class="pill" style="background:#245bea;color:white">DIGITAL TRAVEL PASS</span>
  <h2>Your journey, simplified.</h2>
  <p>Apply for a student or route pass, renew monthly, manage your demo wallet and generate a QR code for verification.</p>
  <button class="primary" onclick="page('apply')">Apply for a Pass →</button>
  <button class="secondary" onclick="page('myPass')">View My Pass</button>
 </div>
 <div class="grid">
  <div class="card"><span class="muted">My applications</span><div class="metric" id="count">0</div></div>
  <div class="card"><span class="muted">Wallet balance</span><div class="metric" id="balance">₹0</div></div>
  <div class="card"><span class="muted">Approved passes</span><div class="metric" id="approved">0</div></div>
 </div>
 <div class="card"><h3>How it works</h3>
  <div class="grid">
   <div><h3>① Apply</h3><p class="muted">Submit your details and select a pass.</p></div>
```

```
  <div><h3>② Approval</h3><p class="muted">Admin reviews the application.</p></div>
  <div><h3>③ Travel</h3><p class="muted">Show your QR code for demo verification.</p></div>
 </div>
 </div>
</section>

<section id="apply" class="hidden">
 <h2>📝 Apply for a Bus Pass</h2>
 <div class="card">
 <form id="applicationForm" onsubmit="applyPass(event)">
  <div class="row">
  <div><label>Full Name</label><input id="name" required placeholder="Enter full name"></div>
  <div><label>Date of Birth</label><input id="dob" type="date" required></div>
 </div>
  <div class="row">
  <div><label>Email</label><input id="email" type="email" required placeholder="name@example.com"></div>
  <div><label>Phone Number</label><input id="phone" type="tel" required pattern="[0-9+() -]{7,16}" placeholder="Phone number"></div>
 </div>
  <label>Address</label><input id="address" required placeholder="Residential address">
  <div class="row">
  <div><label>Pass Category</label>
   <select id="category" onchange="updateFee()">
    <option value="Student">Student Pass</option>
    <option value="Route">Route Pass (Working Person)</option>
```

```
  </select>
  </div>
  <div><label>School / Organization</label><input id="institution" required placeholder="Institution name"></div>
 </div>
 <div class="row">
  <div><label>From</label><select id="from" required></select></div>
  <div><label>To</label><select id="to" required></select></div>
 </div>
 <div class="row">
  <div><label>Pass Duration</label>
   <select id="duration" onchange="updateFee()">
    <option value="1">1 month</option><option value="3">3 months</option><option value="6">6 months</option>
   </select>
  </div>
  <div><label>Demo Fee</label><input id="fee" readonly></div>
 </div>
 <label>Pass Photo (optional)</label><input id="photo" type="file" accept="image/*">
 <p class="muted">Demo pricing: student passes cost ₹100/month and route passes ₹300/month. These are sample prices, not official fares.</p>
 <button class="primary" type="submit">Submit Application</button>
</form>
</div>
</section>

<section id="wallet" class="hidden">
 <h2>💳 My Demo Wallet</h2>
 <div class="grid">
```

```
 <div class="card"><p class="muted">Available balance</p><div class="metric" id="walletBalance">₹0</div></div>
 <div class="card"><h3>Add Money</h3>
  <form onsubmit="addMoney(event)">
   <label>Amount (₹)</label><input id="topup" type="number" min="1" max="10000" required>
   <button class="primary">Add Demo Balance</button>
  </form>
  <p class="muted">No real payment is processed.</p>
 </div>
</div>
<div class="card"><h3>Wallet Transactions</h3><div id="transactions"></div></div>
</section>

<section id="myPass" class="hidden">
 <h2>🎫 My Pass & e-Ticket</h2>
 <div class="card">
  <label>Find Pass by ID</label>
  <div class="row"><input id="passSearch" placeholder="e.g. BP1001">
   <button class="primary" onclick="findPass()">Find Pass</button></div>
  <div id="passResult" style="margin-top:18px"></div>
 </div>
 <div class="card" style="margin-top:16px">
  <h3>Buy Demo e-Ticket</h3>
  <form onsubmit="buyTicket(event)">
   <div class="row"><div><label>Passenger Name</label><input id="ticketName" required></div>
   <div><label>Ticket Fare (₹)</label><input id="ticketFare" type="number" min="1" value="30" required></div></div>
```

```
 <div class="row"><div><label>From</label><select id="ticketFrom"></select></div>
 <div><label>To</label><select id="ticketTo"></select></div></div>
 <button class="primary">Buy e-Ticket</button>
</form>
<div id="ticketResult"></div>
</div>
</section>

<section id="routes" class="hidden">
<h2>📍 Bus Routes & Timings</h2>
<div class="card">
<p class="muted">Sample route information for demonstration.</p>
<div class="tablewrap"><table>
<tr><th>Route ID</th><th>From</th><th>To</th><th>Departure</th><th>Estimated Duration</th></tr>
<tr><td>R101</td><td>Central Station</td><td>City College</td><td>08:00 AM</td><td>25 min</td></tr>
<tr><td>R102</td><td>Central Station</td><td>Tech Park</td><td>08:30 AM</td><td>40 min</td></tr>
<tr><td>R103</td><td>City College</td><td>Tech Park</td><td>09:00 AM</td><td>20 min</td></tr>
<tr><td>R104</td><td>Market</td><td>Central Station</td><td>09:15 AM</td><td>15 min</td></tr>
</table></div>
<label>Calculate Estimated Travel Time</label>
<div class="row"><select id="routeFrom"></select><select id="routeTo"></select></div>
<button class="primary" style="margin-top:12px" onclick="estimateTime()">Calculate Time</button>
<p id="timeResult" class="muted"></p>
```

Sample data only; this is not a live public transport timetable.

</section>

<section id="conductor" class="hidden">

<h2>🧻 Conductor Portal</h2>

This is a simulated QR verification tool. It is not connected to a real scanner, bus GPS or transport authority.

<h3>Verify Pass / Ticket ID</h3>

<form onsubmit="verifyPass(event)">

<label>Enter ID</label><input id="verifyId" required placeholder="BP1001 or ET2001">

<button class="primary">Verify ID</button>

</form>

<h3>Update Demo Bus Location</h3>

<label>Bus Number</label><input id="busNo" placeholder="BUS-101">

<label>Current Stop / Location</label><input id="busLocation" placeholder="Current location">

<button class="primary" onclick="updateBus()">Update Location</button>

</section>

```
<section id="admin" class="hidden">
 <h2>⚙️ Admin Portal</h2>
 <div class="warning">Educational demo: approval and conductor features have no real authentication. Do not use for actual student records or payments.</div>
 <div class="card">
  <h3>Pass Applications</h3>
  <div class="tablewrap" id="adminTable"></div>
 </div>
 <div class="card" style="margin-top:16px">
  <h3>Conductor Account (Sample)</h3>
  <p class="muted">Sample ID: COND-01 · Status: Demo account</p>
  <h3>Bus Information</h3>
  <p id="busInfo" class="muted">No demo location submitted.</p>
  <button class="secondary" onclick="exportReport()">Export Application Report (CSV)</button>
 </div>
</section>
</main>
<footer>SwiftPass · Internet Programming Lab · Browser-based prototype</footer>

<script>
const DB="swiftpass_demo_v1";
const stops=["Central Station","City College","Tech Park","Market","Bus Terminal"];
function load(){
 try{return JSON.parse(localStorage.getItem(DB))||{passes:[],tickets:[],balance:0,transactions:[],bus:null}}
 catch{return {passes:[],tickets:[],balance:0,transactions:[],bus:null}}
}
```

```
let data=load();
function save(){localStorage.setItem(DB,JSON.stringify(data));render()}
function money(n){return "₹"+Number(n).toLocaleString("en-IN")}
function esc(v){return String(v??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;",'"':"&#39;"}[c]))}
function page(id){
 document.querySelectorAll("main section").forEach(s=>s.classList.add("hidden"));
 document.getElementById(id).classList.remove("hidden");render()
}
function fillStops(id){
document.getElementById(id).innerHTML=stops.map(s=>`<option>${esc(s)}</option>`).join("")
}
["from","to","ticketFrom","ticketTo","routeFrom","routeTo"].forEach(fillStops);
function updateFee(){
 const rate=document.getElementById("category").value==="Student"?100:300;
document.getElementById("fee").value=money(rate*Number(document.getElementById("duration").value))
}
updateFee();

function applyPass(e){
 e.preventDefault();
 const photo=document.getElementById("photo").files[0];
 if(photo&&photo.size>1024*1024){alert("Choose a photo under 1 MB.");return}
 const category=document.getElementById("category").value;
 const months=Number(document.getElementById("duration").value);
```

```
 const rate=category==="Student"?100:300;
 const item={
id:"BP"+(1001+data.passes.length),name:document.getElementById("name").value.trim(),
dob:document.getElementById("dob").value,email:document.getElementById("email").value,
phone:document.getElementById("phone").value,address:document.getElementById("address").value,
  category,institution:document.getElementById("institution").value,
from:document.getElementById("from").value,to:document.getElementById("to").value,
  months,fee:rate*months,status:"Pending",created:new Date().toISOString().slice(0,10),
  expiry:new Date(Date.now()+months*30*86400000).toISOString().slice(0,10)
 };
 data.passes.push(item);
 alert("Application submitted! Your Pass ID is "+item.id+". Wait for demo admin approval.");
 e.target.reset();updateFee();save();page("myPass");
 document.getElementById("passSearch").value=item.id;findPass();
}
function addMoney(e){
 e.preventDefault();const amount=Number(document.getElementById("topup").value);
 if(amount<=0||amount>10000)return alert("Enter an amount from ₹1 to ₹10,000.");
 data.balance+=amount;data.transactions.unshift({text:"Demo wallet top-up",amount,date:new Date().toLocaleString()});
 e.target.reset();save();alert("Demo balance added.")
```

```
}
function buyTicket(e){
 e.preventDefault();const fare=Number(document.getElementById("ticketFare").value);
 if(fare<=0)return alert("Enter a valid fare.");
 if(data.balance<fare)return alert("Insufficient demo wallet balance. Add money first.");
 const
t={id:"ET"+(2001+data.tickets.length),name:document.getElementById("ticketName").v
alue,

fare,from:document.getElementById("ticketFrom").value,to:document.getElementById(
"ticketTo").value,
 date:new Date().toLocaleString()};
 data.balance-=fare;data.tickets.push(t);
 data.transactions.unshift({text:"e-Ticket "+t.id,amount:-fare,date:t.date});
 save();
 document.getElementById("ticketResult").innerHTML=`<div class="pass"><h3>e-
Ticket Created</h3><p>ID: <b>${esc(t.id)}</b></p><p>${esc(t.from)} →
${esc(t.to)}</p><p>Fare: ${money(t.fare)}</p><button class="secondary"
onclick="showQR('${t.id}')">Generate Ticket QR</button><div id="qr"></div></div>`;
}
function findPass(){
 const id=document.getElementById("passSearch").value.trim().toUpperCase();
 const p=data.passes.find(x=>x.id.toUpperCase()===id);
 const t=data.tickets.find(x=>x.id.toUpperCase()===id);
 const out=document.getElementById("passResult");
 if(!p&&!t){out.innerHTML="<p class='muted'>No matching pass or ticket. Submit an
application or purchase a ticket first.</p>";return}
 if(t){out.innerHTML=`<div class="pass"><h3>SwiftPass e-Ticket</h3><p>ID:
${esc(t.id)}</p><p>Name: ${esc(t.name)}</p><p>Route: ${esc(t.from)} →
${esc(t.to)}</p><p>Fare: ${money(t.fare)}</p><button class="primary"
onclick="showQR('${t.id}')">Show QR Code</button><div id="qr"></div></div>`;return}
```

```
 out.innerHTML=`<div class="pass"><h3>🚌 SwiftPass ${esc(p.category)}</h3><p><b>Pass ID:</b> ${esc(p.id)}</p><p><b>Name:</b> ${esc(p.name)}</p><p><b>Institution:</b> ${esc(p.institution)}</p><p><b>Route:</b> ${esc(p.from)} → ${esc(p.to)}</p><p><b>Validity ends:</b> ${esc(p.expiry)}</p><p><b>Fee:</b> ${money(p.fee)}</p><p><b>Status:</b> <span class="status">${esc(p.status)}</span></p>${p.status==="Approved"?`<button class="primary" onclick="showQR('${p.id}')">Generate Pass QR</button><button class="secondary" onclick="renew('${p.id}')">Renew Pass</button>`:""}<div id="qr"></div></div>`;

}

function showQR(id){

 const target=document.getElementById("qr");if(!target)return;

 target.innerHTML="";

 if(!window.QRCode){target.innerHTML="<p>QR library unavailable. Check internet and reload.</p>";return}

 new QRCode(target,{text:id,width:180,height:180,colorDark:"#12264b",colorLight:"#ffffff",correctLevel:QRCode.CorrectLevel.M});

}

function renew(id){

 const p=data.passes.find(x=>x.id===id);if(!p)return;

 if(p.status!=="Approved")return alert("Only approved passes can be renewed.");

 const fee=(p.category==="Student"?100:300);

 if(data.balance<fee)return alert("Add enough demo wallet balance to renew.");

 data.balance-=fee;p.fee+=fee;p.expiry=new Date(Date.now()+30*86400000).toISOString().slice(0,10);

 data.transactions.unshift({text:"Renewal "+p.id,amount:-fee,date:new Date().toLocaleString()});

 save();findPass();alert("Demo renewal completed for one month.");

}

function verifyPass(e){
```

```
 e.preventDefault();const id=document.getElementById("verifyId").value.trim().toUpperCase();

 const p=data.passes.find(x=>x.id.toUpperCase()===id);

 const t=data.tickets.find(x=>x.id.toUpperCase()===id);

 const result=document.getElementById("verification");

 if(p&&p.status==="Approved")result.innerHTML=`<div class="pass"><h3>✅ Pass Found</h3><p>${esc(p.name)} · ${esc(p.id)}</p><p>${esc(p.from)} → ${esc(p.to)}</p><p>Valid until ${esc(p.expiry)}</p></div>`;

 else if(t)result.innerHTML=`<div class="pass"><h3>✅ Ticket Found</h3><p>${esc(t.name)} · ${esc(t.id)}</p><p>${esc(t.from)} → ${esc(t.to)}</p></div>`;

 else if(p)result.innerHTML="<p class='warning'>Pass is pending approval; it is not valid for demo verification.</p>";

 else result.innerHTML="<p class='warning'>No matching pass or ticket found.</p>";

}

function updateBus(){

 const no=document.getElementById("busNo").value.trim(),loc=document.getElementById("busLocation").value.trim();

 if(!no||!loc)return alert("Enter bus number and location.");

 data.bus={no,loc,time:new Date().toLocaleString()};save();

 document.getElementById("busStatus").textContent="Demo location updated.";

}

function estimateTime(){

 const a=document.getElementById("routeFrom").value,b=document.getElementById("routeTo").value;

 const minutes=Math.abs(stops.indexOf(a)-stops.indexOf(b))*12;

 document.getElementById("timeResult").textContent=a===b?"Choose two different stops.":"Estimated demo travel time: "+minutes+" minutes. This is not live traffic data.";

}
```

```
function approve(id){
 const p=data.passes.find(x=>x.id===id);if(p){p.status="Approved";save()}
}
function reject(id){
 const p=data.passes.find(x=>x.id===id);if(p){p.status="Rejected";save()}
}
function render(){
 document.getElementById("count").textContent=data.passes.length;

document.getElementById("approved").textContent=data.passes.filter(p=>p.status==="Approved").length;
 document.getElementById("balance").textContent=money(data.balance);
 document.getElementById("walletBalance").textContent=money(data.balance);
 document.getElementById("transactions").innerHTML=data.transactions.length?
 "<table><tr><th>Transaction</th><th>Amount</th><th>Date</th></tr>"+

data.transactions.map(t=>`<tr><td>${esc(t.text)}</td><td>${money(t.amount)}</td><td>${esc(t.date)}</td></tr>`).join("")+"</table>":"<p class='muted'>No transactions yet.</p>";
 document.getElementById("adminTable").innerHTML=data.passes.length?
 "<table><tr><th>ID / Name</th><th>Category</th><th>Route</th><th>Fee</th><th>Status</th><th>Action</th></tr>"+

data.passes.map(p=>`<tr><td>${esc(p.id)}<br>${esc(p.name)}</td><td>${esc(p.category)}</td><td>${esc(p.from)}→${esc(p.to)}</td><td>${money(p.fee)}</td><td>${esc(p.status)}</td><td><button class='secondary' onclick="approve('${p.id}')">Approve</button> <button class='secondary' onclick="reject('${p.id}')">Reject</button></td></tr>`).join("")+"</table>":"<p class='muted'>No applications submitted.</p>";
 document.getElementById("busInfo").textContent=data.bus?`${data.bus.no} — ${data.bus.loc} (updated ${data.bus.time})`:"No demo location submitted.";
```

```
}

function exportReport(){

 const
fields=["id","name","category","institution","from","to","fee","status","created","expiry"];

 const
rows=[fields.join(","),...data.passes.map(p=>fields.map(k=>`"${String(p[k]??"").replace(
/"/g,'""')}"`).join(","))];

 const url=URL.createObjectURL(new Blob([rows.join("\n")],{type:"text/csv"}));

 const a=document.createElement("a");a.href=url;a.download="swiftpass-
report.csv";a.click();URL.revokeObjectURL(url);

}

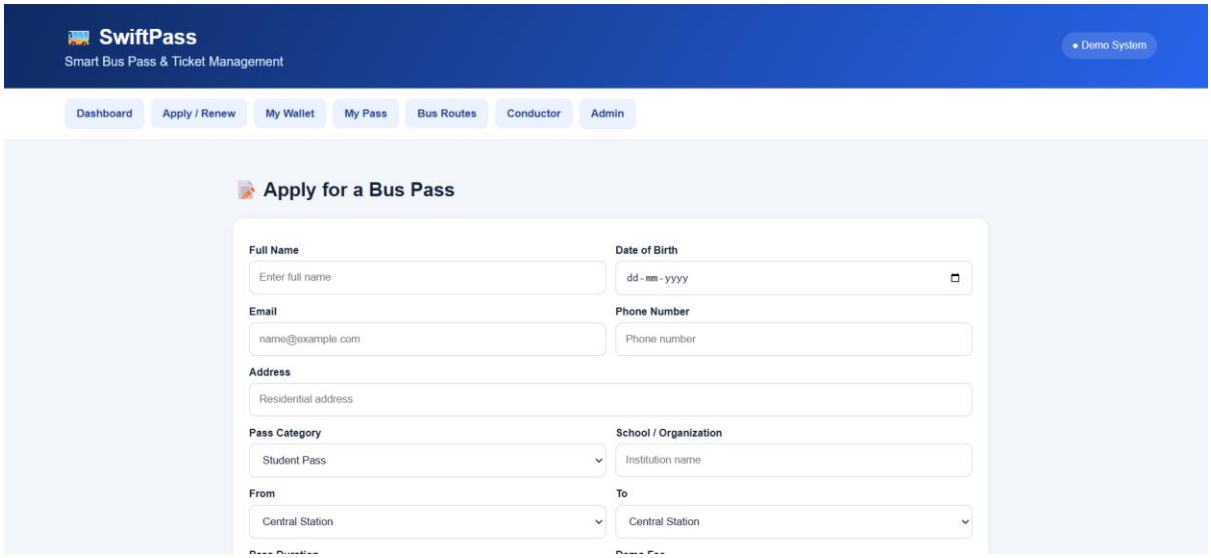
render();

</script>

</body>

</html>
```

Output:



Result

Thus, the Online Bus Pass Management System was implemented as an interactive web prototype using HTML, CSS, and JavaScript. It demonstrates pass application and

renewal, wallet transactions, e-ticket purchases, QR generation, admin approval, and conductor-side verification.